LAB 2: DATABASE DESIGN AND NORMALIZATION
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Part 2: Normalization (1NF to 3NF)

Unnormalized Table:
matric_no | student_name | student_dept | course_codes | course_titles | lecturers | lecturer_depts | session

1. First Normal Form (1NF)
- We removed multi-valued cells so that each cell holds only a single atomic value.
- Each student-course registration pair was given its own separate row.
- We assigned a composite primary key consisting of (matric_no, course_codes).

2. Second Normal Form (2NF)
- We removed partial dependencies where non-key attributes depended on only part of the composite primary key.
- The table was split into separate Student, Course, and Enrolment tables so that student details depend only on matric_no, and course details depend only on course_code.

3. Third Normal Form (3NF)
- We removed transitive dependencies where non-key columns depended on other non-key columns (such as department names and lecturer details).
- We moved Department details and Lecturer details into their own standalone tables and linked them using foreign keys.

Final 3NF Schema

- Department (dept_id PK, dept_name)
- Student (student_id PK, matric_no, full_name, dept_id FK Department)
- Lecturer (lecturer_id PK, lecturer_name, dept_id FK Department)
- Course (course_id PK, course_code, course_title)
- Enrolment (enrolment_id PK, student_id FK Student, course_id FK Course, session)
- Course_Lecturer (assignment_id PK, course_id FK Course, lecturer_id FK Lecturer, session)